

AnyCam 2.6.6 Audit Report

Release audit - 02 October 2026 - motion by pixels, live cards, release checks - tag 2.6.6

2.6.6 replaces file-size motion detection with pixel comparison, gives each camera its own recording settings from a cog on its card, plays live video in the camera cards, fixes four bugs (B4, B5, B9, B10), and completes release-engineering items F1 to F7. Scope confirmed by CrystalHeeler on 2026-09-30, then extended the same day after his first field test (per-camera settings, global override, tuning line, card changes).
camera_discovery.py: 937 lines added, 182 removed; verify_release.py: 204 added, 11 removed.

What 2.6.5 confirmed working

Item	Evidence
Enhanced View	CrystalHeeler, 2026-09-30: landscape full screen, menus removed, loading message, the Oak-D camera camera live, all good.
Cards, button, changelog	Loading text, the Record button's state, and the HA changelog after an update: good.
Recordings stop	Log 2026-09-30: a 9 s clip at 06:21.
Motion loops without a viewer	Log 07:34 to 14:21: both Lorex loops ran all day, one snapshot every 1.9 s.
Not working	Motion on the Lorex channels: nothing recorded after 08:24 at sensitivity 50 (a 5.1% file-size change). The reason for 2.6.6's main change.

Motion detection: design and evidence

File sizes cannot see a person in a 9 to 10 KB snapshot. Each frame is now shrunk to 64 x 48 grey cells and compared with the previous one at most once a second. Three design steps came from measurements, not assumptions:

Step	What the first version measured	Change
Brightness	Cancelling only the mean: a 30% brighter picture read as 7.4% changed, over the 5% default.	Normalise by mean and spread (contrast) as well: now 0%.
Light rule	After normalising, even an unrelated picture read as only 48% changed, so a '75% changed' rule could never fire.	Measure spread instead: change in more than 75% of 16 areas is light. Person block: 25%; inverted picture: 100%.
Scale	CrystalHeeler: 1 to 100 shown, 1% to 74% of the picture hidden, default 5%.	Log mapping; 63 = 5.0%, finer steps at the sensitive end.

Synthetic test (704 x 480 scene)	Changed	Spread
New camera noise only	0%	0%
30% brighter / darker; 40% more contrast	0%	0%
Night-mode style (grey, darker)	0%	0%
Person block, 1.2% of the picture	1.3%	6%

Synthetic test (704 x 480 scene)	Changed	Spread
Person block, 4.7%	5.0%	25%
Whole picture inverted	72%	100% (light)

Cost measured on the build PC: 0.9 ms to decode and 0.4 ms to compare one frame. A Raspberry Pi 4 is slower; at one comparison per second per armed camera the load stays small, but it runs on the event loop (see 1.4 below).

First field test and what changed

CrystalHeeler installed the first 2.6.6 build on 2026-09-30, armed two cameras and walked past one twice for at least 8 s. Nothing recorded. His Motion Sensitivity was 5; on the 1-100 scale that is 62% of the picture. The log showed both loops running and no near-miss line, because near misses were logged only above half the threshold.

He asked for a slider. Home Assistant cannot draw one for an add-on setting: its add-on configuration page maps every integer to a number box (supervisor-app-config.ts, mode "box"), and the Supervisor sends integer ranges as lengthMin/lengthMax, which that page ignores. So the slider moved into AnyCam's own page.

Change	Detail
Cog menu	Per camera: sensitivity slider 1-100 (value shown as it moves), live reading in slider units with an orange marker, cooldown, tail, file length, folder under /media. Saved in /data/motion.json. Defaults 63, 5 s, 3 s, 30 s, /media/anycam.
Global override	global_recording_settings (off): when on, the five Global settings apply to every camera and the panel is read-only; POST and DELETE return 409.
Tuning line	Once a minute per armed camera: largest change, the sensitivity that would have recorded it, comparisons, light changes. INFO when something moved, DEBUG otherwise.
Folder validation	posixpath.normpath, then must be /media or below; '..' that climbs out is refused. Found by the Windows test run: os.path.normpath rewrote '/media/x' with backslashes.
Cards	Protocol, IP and port at the top of Identity; Test Stream button and /stream/{id}/test removed.
Second round (2026-10-01)	Identity opens from an info icon after the name; lock at the end of the button row; cards keep their own height (grid align-items:start), measured: no gap under the buttons, fullest row fits from 413 px; a single-file recording is renamed without _part01; stray closing div removed.

Third round: live detection and a 3 s pre-roll

CrystalHeeler, 2026-10-01: recordings caught only the end of each event, even at sensitivity 80 to 90, and played choppily. Detection compared snapshots that the Lorex channels deliver every 1.9 s, and the recording connected to the camera only after motion was found, then waited for a keyframe. He asked for a fixed 3 s pre-roll on every event.

Part	Design
Detection	_motion_detector: ffmpeg decodes the smallest RTSP stream (Lorex subtype=1; MJPEG allowed) to 64 x 48 grey at 4 frames a second; each frame is compared with the one 1 s before. Cameras with only a stream wider than 1920, or no RTSP, keep snapshots.

Part	Design
Buffer	<code>_motion_buffer</code> : the main stream copied (video unchanged, audio to AAC) into MPEG-TS; <code>_TsBuffer</code> keeps the newest keyframe at least 3 s old and everything after it, found by the TS random-access flag.
Writer	On motion: an ffmpeg reading MPEG-TS from its input gets PAT, PMT and the buffered GOPs, then each live packet; packets that arrive while it starts are queued and handed over in order. Closing its input finishes the file. <code>hvc1</code> tag for H.265.
Cost per armed camera	Two camera connections held open, a light decode, a few MB of memory.

Other changes

Item	Change
C12 recording length	Segment muxer, a new file every 10 s to 5 min (default 1 min), <code>motion_<date>_<time>_partNN.mp4</code> . Recording claims its slot before its first await and stops once.
C1 live cards	<code>/api/go2rtc/card</code> picks the smallest playable stream; Dahua subtype=1 for DVR channels; over 1920 wide stays on snapshots. One player per camera, moved into each redrawn card; pauses off screen, in the background and during Enhanced View.
B4	13 onclick handlers encoded with jsArg (JSON, then HTML escaping); <code>esc()</code> escapes single quotes; info bar and IP badge escaped. The old card-name handler broke on a backslash.
B5	<code>___COMMUNITY___</code> filled as a JSON literal with <code><</code> escaped.
B9	<code>_stop_proc</code> : wait for an exiting ffmpeg before <code>kill()</code> ; Python 3.9+ <code>kill()</code> polls and so reaped it before <code>asyncio</code> 's watcher (CPython 3.11 source read).
B10	<code>go2rtc</code> , then web server, then hardware probe; <code>snap_loop</code> waits for <code>_HW_PROBED</code> ; a card retries if <code>go2rtc</code> is starting.
F1-F7	Gates 6 and 7, gate 3 extended, 7 contracts; <code>build.yaml</code> removed and <code>BUILD_FROM</code> defaulted; 20 bundles to PDFs; <code>CLAUDE.md</code> refreshed.

Tests run

Suite	Result	Scope
Release gate	PASS	Seven gates, 37 contracts, 19 settings in agreement. Gate 7 checked 5 apt pins, 3 pip pins, 2 <code>go2rtc</code> digests and the base image online.
Gate self-test	PASS	Broke six things in a copy: an unknown setting, an unfilled placeholder, a stale apt pin, a wrong digest, <code>async write_bytes</code> , and the HTTP motion call. Each failed the gate.

Suite	Result	Scope
Python integration	PASS	204 of 204, up from 120. Settings API (defaults, save, six refusals, persistence, Defaults, global override), per-camera cooldown and tail, tuning line, file renaming, card layout, the TS buffer (keyframe choice, split reads), and the detector, buffer and writer end to end with stand-in processes: pre-roll from before the motion, no packet lost or repeated at the hand-over. Also: detector (brightness, contrast, spread, scale), real HTTP loop recording on a person, card stream choice for Lorex, Hikvision, Oak-D, B5 placeholder, B9 stop, B10 order and probe wait.
JavaScript behaviour	PASS	106 of 106, up from 75. Cog panel reading and read-only state. Also: New: hostile names through jsArg in a simulated browser; live cards (attach, redraw, fallbacks, retry, watchdog, pause, prune).
Visual check	PASS	Card and cog panel rendered in a browser with stand-in data: cog placement, Identity order, slider drag 63 to 92 updating the value, marker at 72, read-only state.
Not run (2)	NOTED	The detection, buffer and writer ffmpeg commands were checked against the ffmpeg 5.1 documentation and tested with stand-in processes only.
Not run	NOTED	ffmpeg is not installed on the build PC (rule 4): the segment command is checked against the ffmpeg 5.1 documentation only. No Docker: the image was not built.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, as CLAUDE.md rule 1 requires. **Scope change from 2.6.5:** the full document was restored on 2026-09-30 (sections 1.6 to 1.18 and Part 6 had been lost from the repository; CrystalHeeler found the May 2026 copy). This is the first audit since 2.3.2 scored against all six parts. Audits 2.6.0 to 2.6.5 marked such rows *outside Parts 1-5*.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, duplicates, 31 contracts	PASS	178 top-level definitions, no duplicates.
1.2 Syntax	compile()	PASS	Gate 1.
1.4 Async/await	Blocking work on the loop	NOTED	Gate 3 now also catches pathlib I/O and found the OUI write (fixed). Motion decoding (about 1 ms per armed camera per second) runs on the event loop by choice; move to a thread if the Pi shows lag.
1.5 Style: line length	Measured added lines	NOTED	116 of 937 exceed 79 characters: 37 Python (up to 100), 79 in the page script, markup and CSS strings.
1.6 Function design	New helpers	PASS	Small single-purpose helpers (<code>_motion_thumb</code> , <code>_motion_diff</code> , <code>_motion_judge</code> , <code>_stop_proc</code> , <code>_go2rtc_card_source</code>); keyword-only grace and timeout in <code>_stop_proc</code> .
1.7 Mutability	Shared state	NOTED	Motion state still lives in the global <code>_MOTION</code> dict, now with stopping and <code>clip_base</code> keys; build plan E2.
1.10 Exceptions	Narrow catches	PASS	<code>_motion_thumb</code> catches <code>OSError</code> , <code>ValueError</code> and <code>DecompressionBombError</code> ; <code>_stop_proc</code> catches <code>ProcessLookupError</code> and <code>TimeoutError</code> ; replaced two bare 'except Exception: pass' pairs.
1.12 Logging	New lines	PASS	Light changes and triggers at INFO with their percentages; near misses at DEBUG, for tuning.
1.17 Stdlib first	New dependency	NOTED	Pillow added: the standard library cannot decode JPEG. See 6.2.
1.18 Testing	Behavioural tests	NOTED	250 checks, still in the session scratchpad, not the repo (build plan E7).
2.2 JS async	No await in <code>forEach</code>	PASS	Card players use per-camera promises; failures fall back.

Standard	Check applied	Result	Note
3.3 / 5.1 Escaping	Camera data into HTML and JS	PASS	B4: every onclick camera value through jsArg; simulated-browser test with quote, backslash and tag payloads.
4.3 CSS units	Card player	PASS	Absolute inset:0 over the 16:9 card; no new vh units.
6.1 Security: paths	Recording folder from the page	PASS	Validated on the server: POSIX-normalised, must stay under /media; levels, times and lengths range-checked; tested with six bad inputs.
6.1 Security	Boundary validation	PASS	Camera-derived values encoded for their context (B4); the community endpoint JSON-encoded with < escaped (B5); card streams go through the existing go2rtc allowlist; no shell.
6.2 Dependencies	Pillow 12.3.0	PASS	Pinned; wheels for CPython 3.11 aarch64 and x86_64 confirmed by gate 7; reason documented in the Dockerfile.
6.3 Version control	Commits and gate	PASS	Nine build commits, one per stage (one restores a Dockerfile edit a script missed), then the release commit; the gate ran on each; tag after the final gate run.

Known issues carried forward

- **File splitting and the image build are untested** (no ffmpeg or Docker on the build PC). First install is the test (A12).
- **Motion thresholds are tuned on synthetic pictures.** Real camera noise, rain and trees may need a different setting; the tuning line and the cog's live reading show where to set each camera.
- **Live cards fell back to snapshots in Chrome** on the first field test (B17); needs the browser console lines.
- **The Storage tab** opens on the global recordings folder; cameras recording elsewhere show in Home Assistant's Media browser.
- **Choppy recordings** (B18): cause unknown; needs a recorded file. The recording path changed in this round, so test again first.
- **Connections:** each armed DVR channel holds two more connections to the DVR; a DVR's connection limit is unknown.
- Opening the classic view still stops the card's stream (B15).
- Skip Non-Reference Frames breaks H.264 cameras (B3).

Acceptance

- The image builds on the Pi with the new base image.
- An armed Lorex channel records a person walking past, page closed.
- The cog opens each camera's settings; the slider value moves with the drag; the live reading appears after a walk-past; settings survive a restart; the global switch makes the panel read-only.
- A long event produces part01, part02 ... of about the chosen length.
- The log shows light changes ignored at dawn and dusk, with percentages.
- Cards play live; the Hikvision card stays on snapshots.
- No 'Cannot connect' and no 'Unknown child process' lines.
- One Supervisor warning about motion_sensitivity after the upgrade.

Files changed

File	Change
camera_discovery.py	Motion detection, recording, live cards, B4, B5, B9, B10, OUI write. Version 2.6.6.
verify_release.py	Gates 6 and 7, gate 3 P4 and P7, 7 contracts.
config.yaml, run.sh, translations/en.yaml	global_recording_settings added; motion_detect_level and motion_clip_length replace motion_sensitivity; the five recording settings renamed Global; defaults 30s and 5 s; storage-browser description. Version 2.6.6.
Dockerfile, build.yaml	Pillow 12.3.0; BUILD_FROM default; build.yaml deleted.
docs	Best-practices document restored; 20 audit PDFs rebuilt; CLAUDE.md, README and build plan updated.
CHANGELOG.md	2.6.6 entry.